

|          |                                                                                                                                                         |             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|          |                                                                                                                                                         |             |
| 2(a)     | (thermal) energy <u>per</u> (unit) mass (to cause change of state)                                                                                      | <b>B1</b>   |
|          | (energy required to cause/released in) change of state at constant temperature                                                                          | <b>B1</b>   |
| 2(b)(i)  | 1. (work done on/against) the <u>atmosphere</u>                                                                                                         | <b>B1</b>   |
|          | 2. <u>water</u> as it turns from liquid to vapour                                                                                                       | <b>M1</b>   |
|          | as potential energy of <u>molecules</u> increases                                                                                                       | <b>A1</b>   |
|          | <b>or</b>                                                                                                                                               |             |
|          | <u>surroundings</u> as its temperature rises                                                                                                            | <b>(M1)</b> |
|          | as energy is lost/transferred to surroundings                                                                                                           | <b>(A1)</b> |
| 2(b)(ii) | $VI - h = M/t \times L$ (where $h$ = power loss)<br><b>or</b><br>$L = (VI t - Q)/M$ (where $Q$ = energy loss)                                           | <b>C1</b>   |
|          | $(14.2 \times 6.4) - (11.5 \times 5.2) = (9.1 - 5.0) \times L/300$<br><b>or</b><br>$L = [(14.2 \times 6.4) - (11.5 \times 5.2)] \times 300/(9.1 - 5.0)$ | <b>C1</b>   |
|          | $L = 2300 \text{ J g}^{-1}$                                                                                                                             | <b>A1</b>   |